

Guide: Repairing the Cycle Mismatch with an Inverse Path

Collatz proof project — September 5, 2026

What the earlier bound ruled out

The forward bridge search could get stuck after both base orbits reached the trivial cycle. An integer quantity combining slope, clock, and cycle phase could only move in one direction. Adding more forward bridges could not repair the opposite mismatch.

The new route

We can transport convergence backward along a verified orbit segment. The number 47 reaches 2 in 65 shortcut steps. Its auxiliary bridge is $47 \mapsto 425$, and 425 reaches 2 in 38 steps. The combined route is

$$2 \longleftarrow 47 \mapsto 425 \longrightarrow 2.$$

Reading the first segment backward makes this a different operation from the forward paths covered by the earlier bound. Its net change in the cycle quantity is negative seven. Lean checks the exact orbit computations and their extension to whole arithmetic progressions.

The premise is still required

The middle bridge invokes transfer at a smaller parameter. Lean retains that premise explicitly. The numerical fact that 47 and 425 converge does not by itself prove convergence for every point on their lifted progressions. The construction checks that each recursive parameter is smaller for every quotient, using enough waiting steps when needed.

Why two excursions are enough in principle

The previous cycle excursion changes the quantity by positive one. The new one changes it by negative seven. Combining them can correct any integer mismatch. This gives a construction for a fixed parameter above 15 when both of its base partners are already known to converge.

Moving backward also changes the clock used to define the arithmetic progression. The binary modulus must retain enough precision for the largest clock ever visited, including clocks before a backward move. The program records and checks this condition.

What was tested

The constructor succeeded for every parameter from 16 through 1000: 985 conditional certificates, with maximum final depth 412. Tests replay all of them at three quotients, including every reverse segment and every smaller-parameter comparison. The exact inverse macro is checked in Lean; the general scheduling argument and the entire census are not yet Lean coverage theorems.

Why this still does not prove Collatz

The construction needs the base target to reach the cycle before it can measure the mismatch. Applying it to every parameter would therefore require the convergence statement we are trying to prove. The inverse route fixes a limitation of the certificate language; it does not remove this global coverage gap.

Verification files

Build `Collatz.InverseCycleBridge` from `lean/`. Run `python3 analysis/inverse_cycle_bridge.py` and `python3 -m unittest discover -s analysis -p test_inverse_cycle_bridge.py`.